MSX and SPECTRAVIDEO

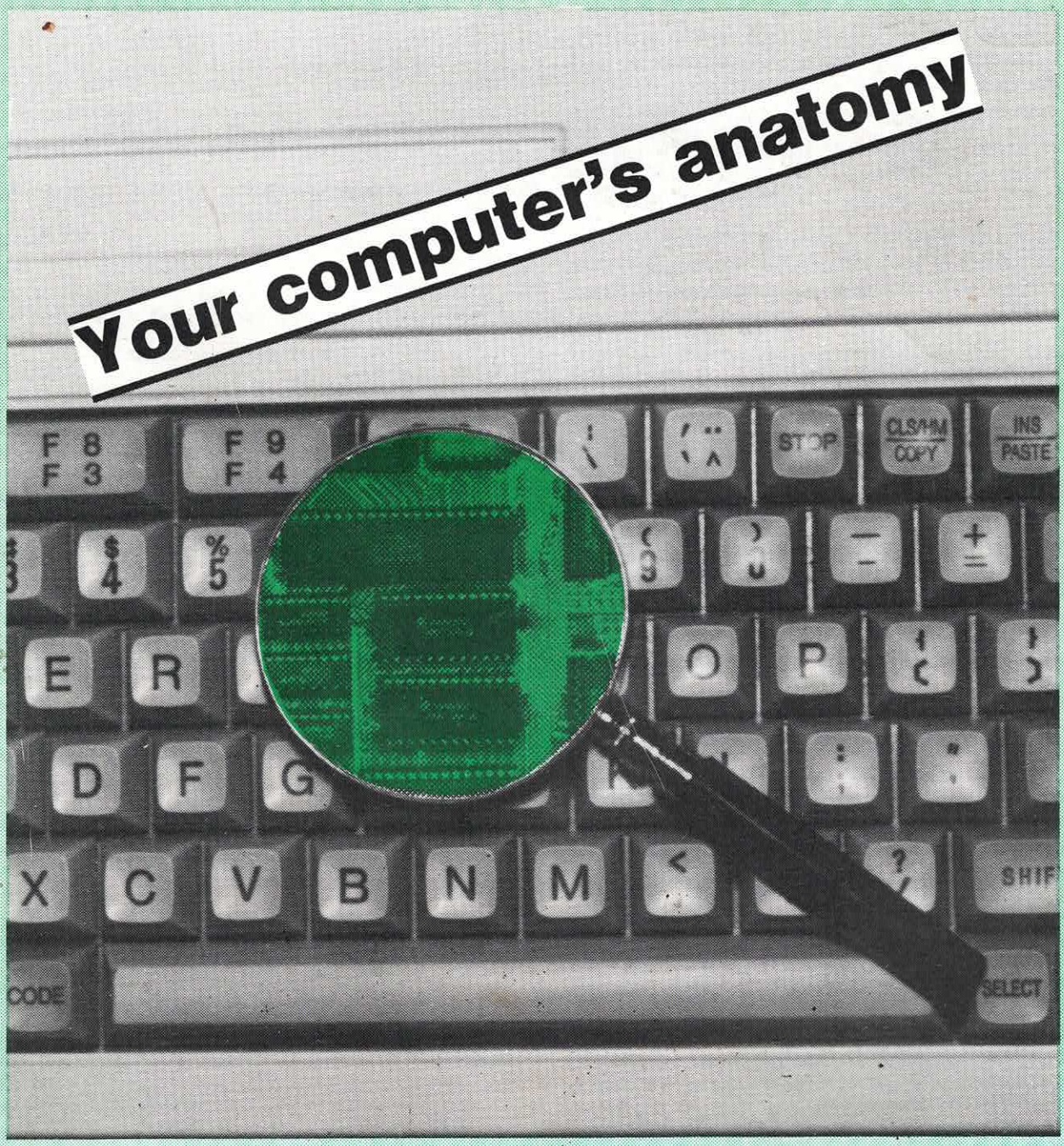
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PETER J FOX (COMPUTERS)



9 ELLEN ST FREMANTLE WA 6160
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MSX and SPECTRAVIDEO
COMPUTER FORUM
Magazine

Volume 1 No. 4

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EDITORIAL

You have probably noticed that Computer Forum is a bit late this month. Due to end-of-month problems we decided to move our production back a couple of weeks.

We have been receiving a great deal of mail from readers expressing support for our magazine. Some wrote about their concern for the future of MSX in Australia. In fact, one lady wondered why the heck we would produce a magazine for such computers when all she has heard is gloom and doom. Well, the answer is simple — we happen to believe in these computers.

What is required now is to convince the Australian distributors of the fact that they are in the home computer industry and to get off their .....s and get the ball rolling.

Computer Forum is trying to do it's bit toward helping MSX take over the market from Amstrad, Commodore etc. We cannot do that without advertising. To advertise in Computer Forum has a two-fold effect. The funds allow us to produce a better magazine, and the advertisements themselves show strength of conviction on the part of the distributor.

You can do your bit too, as this lady did. Write to the Australian distributor of your computer, saying that you read Computer Forum (the ONLY Australian MSX magazine) and show your disappointment at their attitude to marketing your computer in Australia.

Happy computing ...

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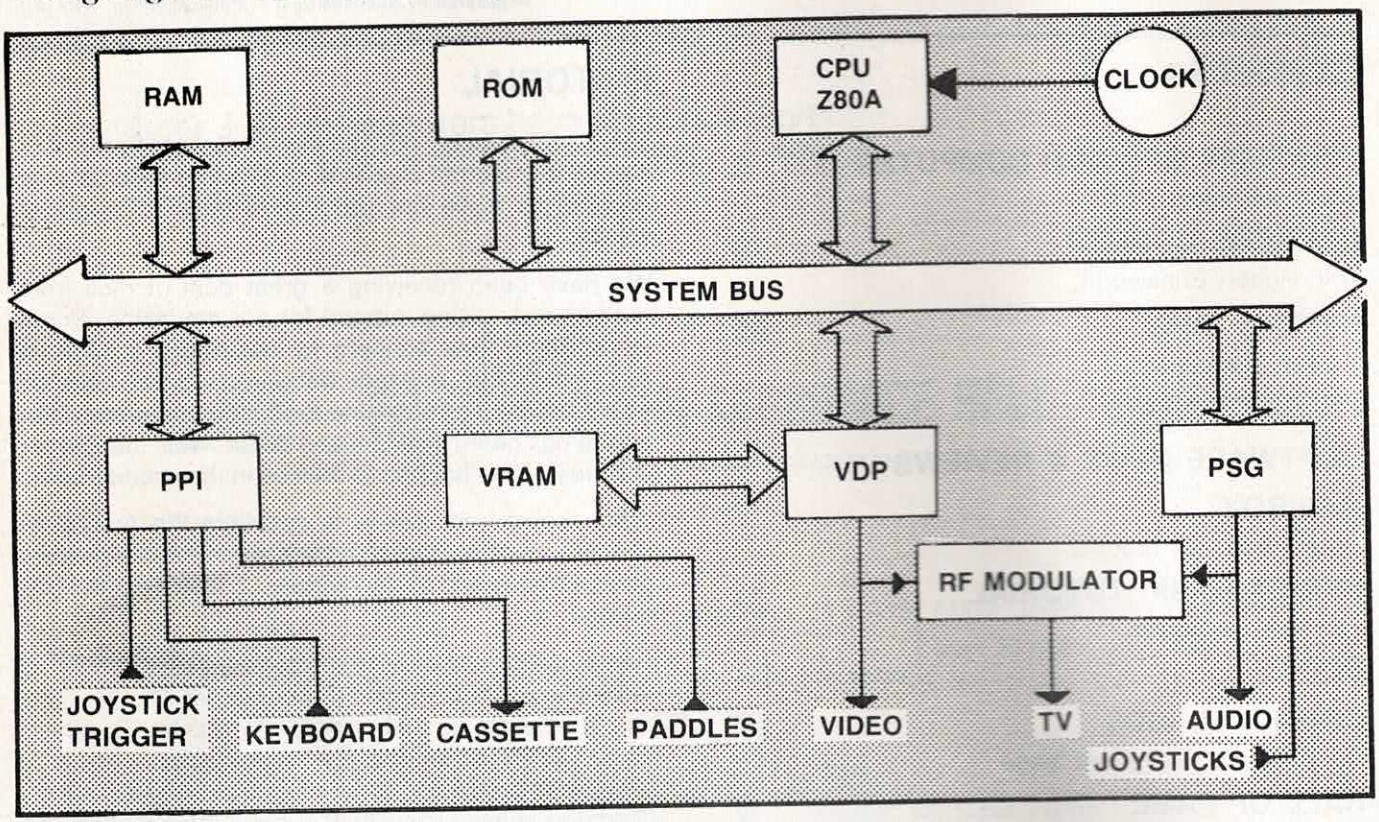
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YOUR COMPUTER'S ANATOMY

The Spectravideo and MSX computers have one thing in common — they all have the same anatomy. That is to say that they all use the same major Chips or Integrated circuits. This article explains how these chips are used to generate the picture, store information and control the "goings-on" inside your computer.



This is a schematic diagram or overview of the main chips used in Spectravideo and MSX computers.

Firstly, let me explain some of the names and terms used in the schematic diagram.

CPU — Central Processing Unit
ROM — Read Only Memory
RAM — Random Access Memory
VRAM — Video RAM
VDP — Video Display Processor
PSG — Programmable Sound Generator
PPI — Programmable Peripheral Interface
RF — Radio Frequency

THE SYSTEM BUS

The term BUS is derived from the bus that people catch to move from one place to another. The System Bus carries signals or electrical pulses to various parts of the computer. The socket on the back of your computer is part of this bus to allow other pieces of equipment to receive these signals.

The signals carried on the bus can be divided into four groups — Address lines, Data lines, Control lines and other signals such as voltages and ground.

THE READ ONLY MEMORY (ROM)

This is the Read Only Memory that contains the BASIC language as a program. Yes, the ROM contains a machine code program which looks at the instructions in the program that you enter or load, and converts it into instructions for the CPU. In the diagram, it is shown having a two-way arrow to the System Bus. This does not mean that you can write information to the ROM. It is there to indicate that the CPU sends a message to the ROM to ask for information.

THE CENTRAL PROCESSING UNIT (CPU)

The type of CPU chip used in these computers is a Z80A. The Z stands for the manufacturer — ZILOG, the 80 indicates that it has 8 data lines and is an 8-bit processor, and the A indicates the speed at which it can process instructions. The Z80A is approximately twice as fast as the Z80 and the Z80B is about three times as fast as the Z80.

The CPU can be regarded as the 'brains' of the computer. It is controlled by the ROM, takes its instructions from the ROM, and controls the flow and timing of instructions throughout the computer.

It is connected to the System Bus by a two-way arrow to indicate that it sends and receives information to and from other parts of the computer.

THE CLOCK

If the CPU is the brains of the computer, then the clock is the heartbeat. This is a silicon crystal which provides pulses at a set rate per second to control the timing of the operations carried out by the computer. Most of the electrical signals within the computer are said to be 'clocked', that is, they must happen at certain times for a certain length of time.

THE RANDOM ACCESS MEMORY (RAM)

The RAM chips can be considered as storage boxes for information which you supply for controlling the sequence of operations carried out by the computer. Your program is stored here. Also in RAM is an area set aside for the computer itself to use as storage for such information as screen control, input/output status etc which is updated by the computer.

THE VIDEO DISPLAY PROCESSOR (VDP)

The VDP is connected to the system bus to receive information from the CPU and supply details of the status of the screen display. The VDP supplies signals directly to the video input of a monitor or mixes with the sound in the RF Modulator for a TV.

THE VIDEO RAM (VRAM)

The VRAM is used by the VDP as a memory bank for the screen. The VDP and VRAM can be considered a small slave computer whose only function is to control the video output.

THE PROGRAMMABLE SOUND GENERATOR (PSG)

The PSG is connected to the system bus to receive information from the CPU and convert this to signals which are then sent directly to the sound input of a monitor or mixed with the video signal in the RF Modulator for sending to the TV input.

The PSG also handles information supplied by the joysticks for transfer to the system bus. The PSG is also a slave processor which is programmed by the main computer. Once the program has been transferred to it, the PSG goes ahead and processes the information without requiring further direction from the main CPU. This is why you can have a PLAY command followed by an INPUT command and the computer will show that it is waiting for input even before the music has finished.

THE PROGRAMMABLE PERIPHERAL INTERFACE (PPI)

The PPI controls the information coming from and going to such peripheral equipment as the joystick trigger, the keyboard, cassette and paddles.

PROGRAM FLOW

Let us have a look at how a typical BASIC program is handled through the different components of the computer. This is a much simplified account of what actually happens. Suppose our program looked like —

```
10 A=STICK(0)
20 PRINT A
```

When the RUN command is entered, the first instruction of the program is scanned for syntax to ensure that the instruction can be performed successfully. If all is well, the first command is read. The CPU asks the BASIC ROM what the heck it all means and it replies with the instructions to be performed for the STICK command.

These instructions tell the CPU to send a request to the PSG for information about the position of joystick number 0. When this information is received by the CPU, it must then examine the System area of RAM to find out the whereabouts in RAM that the variable A is keeping its information. The CPU then sends the position in RAM along the system bus address lines followed by the information along the data lines to be stored in that location in the RAM.

The next instruction of our program is examined for syntax and if all is well, the CPU then asks the BASIC ROM what functions are to be performed to complete that instruction. The ROM says to check with the VDP to

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Your computer's anatomy

ensure that it is alright to display the value of A on the screen. If all is well, then the CPU goes to the System RAM area to find out where the value for A is stored in memory. It then sends that address along the address lines of the system bus, collects the value at that location from RAM transports it along the system bus to the VDP which then stores it in the VRAM at the current cursor position on the screen. The information is then converted to a displayable character and sent via the VDP to the video or TV outlet.

COMPLEX TASK

This may seem like a fairly complex task already. However, the CPU has other tasks to perform at the same time as it is carrying out the instructions that we have given it. These tasks include checking through the RAM and refreshing the information it contains, check the keyboard to see if the user has interrupted the program and remembering the last line number used etc. All of this is done in the time it takes from when you enter RUN to the time that the value of A is displayed on the screen.

How can it do it so fast? Well, remember the clock. It is putting out pulses to the CPU at the rate of 3.58 MILLION PER SECOND and the CPU is handling all of its operations at a similar rate. The rate is not exactly the same, as some instructions must be performed over a period of clock pulses.

Purists Please Note: The above description has been over simplified for the sake of clarity.

So there we have it. At first the computer would appear to be a very intelligent machine, but when you analyse it, you find that it is just a very fast dumb instrument which awaits your commands. It is relying on you to provide the correct instructions. If you do not give the correct instructions, it will quite happily carry out your wrong instructions — AT SEVERAL MILLION OPERATIONS PER SECOND.

Software NEWS & REVIEWS

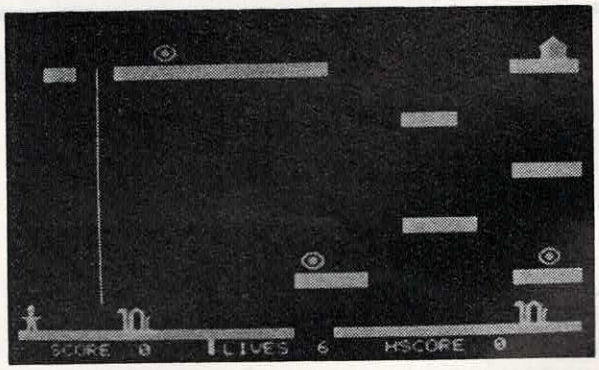


NEW FDS SOFTWARE (MSX & SV)

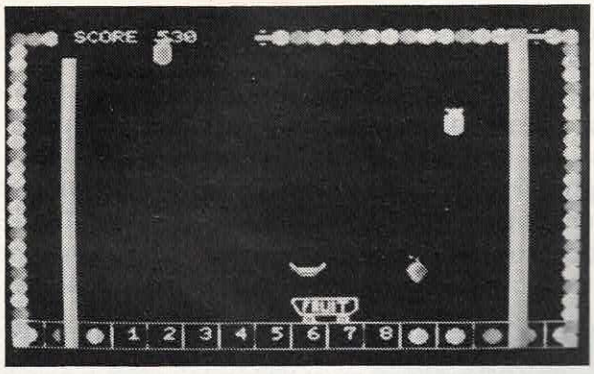
SVPAK26/MSXPAK26

This pack has three titles on the tape — SUPER SMASHER, JUMPING COWBOY, FRUIT CATCHER.

Super Smasher — A breakout type program, you can select from three difficulty levels. You must move the bat to smash the ball against two thicknesses of wall until you breakthrough.



Jumping Cowboy — This is a 'platform' game with nine screens in which you have to collect the opals and go to the brown house on the top platform to get to the next screen.

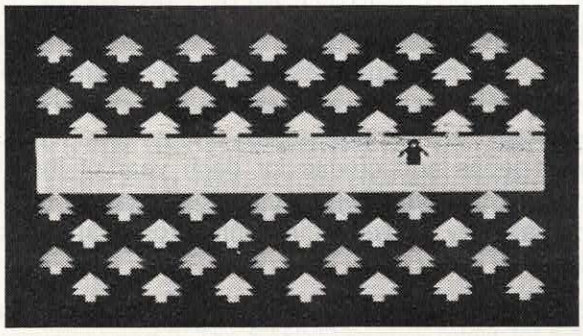


Fruit Catcher — You have to move the cart across the screen and catch the falling fruit. As you progress, the types of fruit increase and fall at different rates.

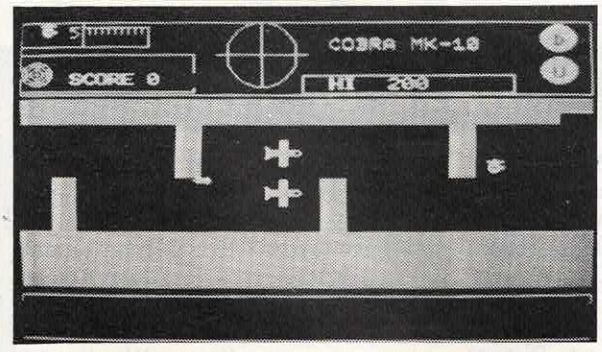
Publisher: Forrest Data Services
Recommended price $15.00

SVPAK27/MSXPAK27

This pack has two titles on the tape — MIRKWOOD FOREST and CHOPPER PILOT.



Mirkwood Forest — A graphics adventure game where you must escape from the forest. To do this you must avoid the spiders, find all four keys, find food and work out how to cross the river.



Chopper Pilot — Guide your helicopter through the canyon and do battle with various enemy fighters, bombers, missiles and UFO's. There are fourteen levels with different combinations of these enemy craft.

Publisher: Forrest Data Services
Recommended price: $15.00

SPECTRAVIDEO and MSX software **Specialists!**

ESVAY - **the** computer version of Yatzhee for hours of entertainment * 1
WORDS - two family type, word testing games with over 200 words! * 1
MEMORY - two memory training programs; Takistoscope and Memory * 1
PERFECT DRAWING - a very good graphics utility for all applications * 1
MEMORY EXPANSION - a program that increases the user RAM to 60K *
BILL MANAGEMENT - an easy to use personal accounts program * 1 3
rated as: "very good, flexible, simple to understand and user friendly"

UTIL 1 - two tape based utility programs: remspc & scrdmp * 1
UTIL 2 - two disk utilities: newdsk & llist; 2 very useful programs 1
MT-BASE - a user friendly ROM based database with many features 2
"recommended whole heartedly" by Canadian Users Group.

AEROBAT - a 3D graphics Cessna 152 flight simulator; does aerobatics!
1986 NZ Software Awards finalist!! * 3
FLIGHT SIMULATOR - fly an airliner, local or across the Tasman * 1
1986 TAX RETURN - let the computer calculate out your tax.(NZ only) * 1
THE MAGIC OF SVI - book & tape for the advanced SVI user ($49.95)
LEUCOCYTE - battle viruses etc in this pacman style game * 3
SEA BATTLE - destroy your opponent's battle ships * 3

* - SVI&MSX tape 1 - SVI disk 2 - MSX cartridge ($99.00) 3 - 3.5" disk
Cassette NZ$29.95; 3.5" disk NZ$39.95; 5.25" disk NZ$34.95
Postage and handling add NZ $5.00 per order. (NZ addresses $1.50)
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May Top 5
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2. Flight Simulator
3. Aerobat
4. ESVAY
5. Bill Management

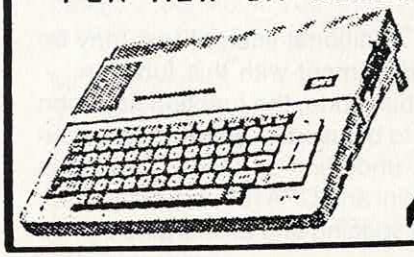
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Software NEWS & REVIEWS

HOME OFFICE (SVI & MSX)
Disk or Cassette

Home Office is a set of three programs designed for the SVI and MSX group of computers. As the name implies, the programs are designed to provide a low-cost Word Processor, Data Base and Accounting Package for home use. Many of the features of the more expensive professional versions are available, within the limitations of the computer's memory capacity and the operating speed of Basic. Separate tape or disk versions are now available. The programs supplied with tape are described below. The disk versions are essentially the same except for loading and saving being disk related and the disk supplied is set up with a menu allowing the programs to be readily selected.

WRITER — the Word Processor.
FILER — the Data Base used to store, retrieve and print records of any type selected by the user.
CHEQUE — a cheque account program which allows records of a cheque account to be maintained. It has facilities for searching and summarizing the account details to aid in budgets.
START — used to set up an empty file format for FILER.
SWR — is a shorter version of WRITER to allow longer documents to be printed.

Also on the tape are WRITE, FILE and CHQ318. These are shorter versions of WRITER, FILER and CHEQUE, with certain functions omitted to allow greater data space on the SV318. The full versions are on the tape for small files or machines with SV328 memory capacity.

WRITER FUNCTIONS

The Writer word processor program has the following functions.

Document Entry — Documents can be entered with word wrap facility and maximum line length specified.
Document Display — The document currently in memory can be viewed with a set number of lines being displayed per screen.
Document Edit — This mode allows you to edit a document by inserting or deleting a character, words or lines.
Append Document — Additional lines of text may be added to an existing document with this function.
Print Document — In this mode, the function keys can select the type of print to be used — emphasised, double width, condensed, underline or italics. Two types of printer, the Star Gemini and CPA100, are supported. Number of copies, line spacing and all margins can be set from this mode.
Save Document — The document can be saved to tape or disk.
Load Document — Files can be loaded from tape or disk as complete documents or appended to an existing one.
Global Change — This allows changing all occurences of one string (OLD STRING) to another (NEW STRING).
Block Move — This allows blocks of up to twenty lines to be moved from one place in the text to another.

FILER FUNCTIONS

The START program must be run to create a new file for use with FILER. This asks for the number of fields per record and the name and length of each field. Once this is entered, the maximum number of records allowed will be displayed.

The FILER program itself allows for data entry, display of a complete file, search facility, alter/delete, sort, save and load.

CHEQUE FUNCTIONS

The cheque account management program allows for entry of cheques, deposits and withdrawals. From the main menu you can select to list cheques, deposits and withdrawals, list all entries, list balances, verify or delete entries, print hard copy, search for type of entry, print monthly summary, load or save a file. The monthly summaries are very comprehensive with a formatted output of debits and credits for each year of the file records. The monthly and annual totals are printed for each category as well as the annual debit and credit amounts.

Lotus 1-2-3 it is not, but at this price — who cares!

Home Office is only available from Downs Home Computing, Toowoomba QLD. Prices are $30.00 for tape and $35.00 for disk. $2.00 postage extra. The individual programs are available separately, however it is cheaper to buy the package.

HALL OF FAME

GAME	SCORE	LEVEL	ACHIEVER
Ali Baba	24650		John Knight
Athletic Land	25200	23	John Knight
Beam Rider	47910	18	John Knight
Disc Warrior	32950		John Knight
Exa Innova (EI)	24640		John Knight
H.E.R.O.	51235	11	John Knight
Lode Runner		49	John Knight
Mouser	35900	6	John Knight
Ninja	23200	1	Peter Beil
River Raid	17760	12	John Knight
Super Crossforce	1307200	1	Peter Beil
Turboat	31360	1	Peter Beil
Zaxxon	52200		John Knight

Congratulations!

PRODUCT NEWS ...



MJY002 MSX
JOY STICK

SMP30 COLOUR
GRAPHIC PRINTER



SANYO JOYSTICK

What can one say about a joystick. It sits there — you can move the lever and press the trigger. Well the SANYO MJY002 joystick does that and more.

Firstly, it has four suction feet which provide 'one hand operation'. The sensitivity of the joystick movement can only be experienced — this one is very good. The movement is smooth with no harsh scratching or sticking. We hope to bring you a small program which might help in assessing the comparitive performance of joysticks.

This is a true MSX joystick having two fire buttons. The main trigger is on the top of the stick and the secondary trigger is in the base. Your MSX and SV computers can recognise two triggers for each joystick. STRIG(1) & STRIG(3) relate to the triggers on the joystick attached to joystick port 1 and STRIG(2) & STRIG(4) for the joystick attached to port 2. The SANYO joystick retails for $20 to $25 depending on where you shop.

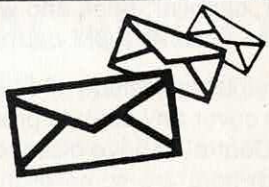
OTHER SANYO PRODUCTS

For a couple of weeks, we also had the use of a SANYO 34cm Colour TV/Monitor and their Colour Graphic Printer. The TV/Monitor was a joy to use. As well as being able to use it as a normal 'box' it has sockets on the back for direct video and audio input. This allows you to connect it to your computer via the TV output or video output. We had hoped to produce photographs of the screen showing the difference in graphics resolution between the two methods. However our camera was not able to record the difference as well as our eyes did. The resolution was almost as good as dedicated colour monitors which sell for twice the price — and you get a TV to boot.

The SMP30 Colour Graphics Printer uses a 4 inch roll of paper and four coloured pens to produce text and graphics from an MSX computer. Unfortunately, SANYO Australia had received a batch of printers which were bound for Europe and the manual was in French, so we were unable to put it through its paces. However, the printer was used by a member of the MSX User Group (WA) who was familiar with this type of printer and it seemed to perform some terrific functions in the high resolution department as well as text. The only disappointment being the width of the paper.

We thank SANYO for their assistance in making this equipment available to us. At time of going to press, they were unable to supply a list of stockists, however, I am sure they would not mind you contacting their office in your state for further information.

MAILBOX



This page is for your comments and queries. If you have a specific problem or have solved one, we would like to hear about it.

If you require an answer before we publish it, please enclose a stamped self-addressed envelope.

DISK-BASED WORD PROCESSOR

I read in MSX and Spectravideo Computer Forum Volume 1, Number 2, a letter from Gregg Faulkner, Weston ACT, wanting to know of Disk based Public Domain word processing programmes.

I am the librarian of the MSX & Spectravideo Computer Club in Auckland and we have just the thing Gregg needs. It's called SVDO.COM and is disk based 80 column and what's more its (Free?)

We don't try to make any money out of our software but we do charge for the disks $12 plus postage and packing $3, so if you could put this letter in your magazine, we may be of help to others.

Mr J. Cammell
39 Strid Road
Te Atata, Auckland 1208
New Zealand

We assume that the amounts quoted are in New Zealand dollars.

MSX 1 AND 2 CONFUSION

I don't know what the world is coming to! Most of us own, or have just bought an MSX computer, which is still rather new, but pushing MSX comes MSXII. I don't know whether I want to buy peripherals and programs for my computer, as they might not work with MSXII, as I might sell, trade-in (whatever) my existing computer. So could you publish in the near future the answers to the following questions —

1. Will MSXI programs run on MSXII computers?
2. Will MSXII peripherals run on MSXI computers and vice-versa?
3. Is MSXII disk BASIC (printer BASIC etc) better, and different to MSXI peripherals existing now?
4. What companies sell, or are going to sell in the near future MSXII computers?
5. Is there any way that I can trade in my MSXI computer, (or get most of my money back) for an MSXII computer?

I would be very grateful, and it would put my mind to rest if you would publish these answers real soon. Could you also publish the memory address of the Caps latch, so it would be possible in a program to turn on and off the Caps LED (for warnings, gimmicks etc.)

Is "Hackers Corner" for MSX. I'm not sure on this but it seems the section is only for Spectravideo and this would disable the magazine somewhat.

In the issues to come could you please include rough prices (if possible) for products in Product News, Software News and Reviews and Books sections.

In the section Software News and Reviews could you give ratings out of five on reviewed programs for graphics, sound quality compared to other similar programs, lasting interest, value for money and state the publisher, supplier, price and whether the program is in cassette, disk or ROM cartridges.

Could you please include another section in Computer Forum to cover any sounds produced by MSX, called "Sound Centre". I have discovered a few sounds and (enclosed) here are some of them.

Paul R. Rode
Curlewis, N.S.W.

We have taken your advice Paul and the sound routines that you submitted have been used to "kick-off" our SOUND CENTRE section elsewhere in this issue. Anybody have more routines like this?

Everything that we have read or been advised about MSXII indicates that MSXI is upwardly compatible with it. We have also read that the MSX software developers guide states that all addressing of the video chip be done only through a certain address location which allows it to check if a "MSXII adaptor" is connected. Make of that what you will — we are still checking, and will keep you, and other readers who are in the same boat, informed.

One of the biggest problems associated with producing Computer Forum, is maintaining a balance between SV and MSX information. In future we will indicate which articles are for either/both machines. With regards prices of software and equipment reviewed in this magazine, we endeavour to publish these if the information is available at time of publication.

We have been reluctant to apply a points system to our evaluation of software, as these would only be our opinion and others may not agree — "You said it had excellent sound. I think it's crummy!" What we try to do in the review is give sufficient information for readers to decide whether to purchase or even look at a piece of software. After all, software is like a novel — you don't know whether you would enjoy it until you have read it.

CAPS LAMP FLASH (MSX VERSION)

```
10 OUT 170,64:OUT 170,0
20 FOR T=1 TO 100: NEXT
30 OUT 170,64:OUT 170,64
40 FOR T=1 TO 100:NEXT
50 GOTO 10
```

CAPS LAMP FLASH (SV VERSION)
Change line 10 to read — 10 OUT 140,0
and line 30 to read — 30 OUT 140,255

PROGRAMMING PROBLEM

Firstly let me thank you on an excellent magazine. It's about time we MSX users have something we can relate to. Please don't let publication stop on this, our first decent users mag.

Now for my problem. I wrote a short program which is used to store our customers repair records. This has worked to some degree with the only real problem being that the information had to be input in data statements. As we are working with six variables, the chances of leaving out the separator comma is very great. Is there another way of handling this task? The records are updated on a daily basis. My manager has also asked me to include a counter in which he can, at the end of the month, determine productivity with regard to the workers initials. We are using a Toshiba MSX and have recently acquired an MSX.DOS disk. Could you tell me what and how this is used please. I have read the manual through three times and can't make head or tail of it.

John B. Reynolds
Windsor, QLD

The first rule of programming is to keep users out of your program by providing a method of inserting or recalling information without having to change anything in the program.

To this end, we have published in our Program Listings for this month, a data entry and retrieval program for Spectravideo and MSX. You should be able to modify this for your own purposes. As for the productivity angle, you could loop through each record at the end of the month, counting the number of entries containing a certain initial.

With regard the MSX.DOS system, we hope to include details of its usage over the coming months.

MACHINE CODE GAMES

First I would like to say what an excellent magazine you have published. I was frustrated in owning what I call an excellent computer but not being able to obtain good games and other software. I own a 328, and I read about the article on machine code games.

I read you mentioned Ninja, Tetra Horror, T. Bunnie etc. Well I have them all. I think more software companies should write software for the 328. I know FDS writes software and they're excellent. So why not write some machine code software?

Secondly I would like to say, that recently I bought a disk drive and I would like to know how I can get my hands on Pascal or Fortran for my 328. It would be most appreciated.

Steven Dimitrio & Mark Bervanakis
Footscray, Victoria.

Machine code software has not been readily available for the Spectravideo due to the secretive nature of the manufacturers regarding memory locations etc. We are hoping to overcome this situation through these pages and, hopefully, some machine code programmers may come forward and produce the goods.

I know that Turbo Pascal is available for the 328 but am not sure about Fortran. I suggest you contact some of the dealers who advertise in Computer Forum.

X'PRESS DISKS

I was interested to see in Vol. 1, No. 1 that you have a 5¼" disk drive attached to the disk port of your SV1738. I do too, just in case I want to buy CP/M software ...!

Also potentially of interest (to MSX-DOS users) is the ability to read IBM PC-DOS (MS-DOS) files from 5¼" disks. This is straightforward on the SV1728 with its SV1707 disk drive unit, or on the SV1738 with a 707 plugged into its cartridge slot. Not, however, with a 5¼" drive plugged into the 738's disk port, since the BIOS routines assume a single-sided 80 track drive is attached.

It would seem to be a relatively simple matter to write a utility to do this, for someone with the appropriate skills, and knowledge of disk mapping and MSX I/O methods. Can any reader help?

By the way, have you tried the SV1707 on the X'press? Due to the slot encoding and order of the boot-up's slot checking, the 707 takes precedence and becomes drives A & B (B could be a second physical drive daisy-chained on the 707's bus). The 707 also appears as drives C & D due to incomplete decoding of the slot address bits; and the computer's internal and external 3½" drives become E & F. MSX-DOS off the 707-supplied 5¼" disk works fine, CP/M does not.

Peter Freeman
Keysborough, Victoria.

Thanks for the low down on the disk drive situation Peter. Can anyone else supply details of what goes with what on MSX? Remember that MSX is selling on the basis of compatibility between equipment from different manufacturers.

CHESS PROGRAM AVAILABLE

Some comments for Mailbox from Vol. 1 No. 3. For Noel Bergman, there is a Chess program available from TJ Software for the SV at $22.50. Unfortunately for the 318, 32K is required.

For Peter Beil's letter, I have a Turbo module running on my coleco adaptor, although mine is second hand and has been converted for use with disk. When using it connected to the SV joystick port, using the stick command, I cannot get any readings from it. I hope this is of some use.

Todd Davies
TJ Software
P.O. Box 164
Donnybrook, W.A.

Todd is a very active MSX/SV user and supplier and is very supportive of this magazine. Thanks' Todd.

LEARN MACHINE CODE

WITH ROBERT BRINKWORTH

Coming up as a separate part of this magazine will be a Machine Language Tutorial which hopefully you will understand without having a degree in computer science. The tutorials will cover —

What is programming?
Flowcharting
Information representation
Basic programming techniques
Addressing techniques
Input/Output techniques
Data structures
Handy hints
Programming for sound and graphics
Library of routines
Assemblers, Disassemblers — what do they all mean?
A simple machine code editor
with more sections added as time progresses.

HACKERS CORNER

Hi. Robert Brinkworth here again with the column that beats all columns. In this issue I will be explaining all about the Video Display Processor (or VDP), also getting to grips with the DOS commands as we look at what must be executed for each command.

NEWS

Recently I have been dis-assembling the game TURBOAT which is full of nice little routines. I hope to include some of these in my Little Disassemblys section in the future. Trying to kill my printer, I set about printing the entire Spectravideo ROM including labels. From this I base all my information which is released to you.

A program which might be interesting to you is my MSX to Spectravideo machine language converter. Although this program is still in the developing stage it has already proved useful in converting PITFALL II, which worked without a hitch after I found all the ports involved in the program.

THE DOS — HOW TO DO THE COMMANDS

Note that the word "parse" means to check the inputted data to see if it is valid or not. APPEND, ATTRIB, AUTO, COPY, DIR, FORMAT, FREE, PDRIVE, PROT, PURGE, R, RENAME, ROUTE, SAVE, STMT, SYSTEM, TIME and VERIFY will be included in the next issue.

AUDIO:
1 parse for ON/OFF
2 set bit needed
3 exit

BASIC2:
1 reset pointers
2 switch memory banks
3 set stack to 0000
4 jp 0000

BEEP:
1 setup PSG for tone delay of 1 second
2 exit

BLINK:
1 parse input
2 set blink flag
3 exit

BOOT:
1 jump to dos entry (cold start)

CHAIN:
1 check if CHNON flag is set
2 read command from disk file
3 increment counters
4 jump do-command

CHNON:
1 parse on or off
2 change console device to disk file or vice-versa
3 exit

CLEAR:
* (parse for start, end and length)
* optional
1 do block move operation
2 exit

CLICK:
1 parse for on or off
2 set flag on value of parse
3 exit

CLOCK:
1 parse input
2 set clock display flag
3 exit

CLS:
1 reset the VDP to text mode (or clear the screen of the 80 column card)
2 exit

COLOR:
1 parse for letters, background
2 set VDP register accordingly
3 exit

CREATE:
1 parse filename
2 make file
3 exit

DATE:
1 parse date input
2 echo date flags
3 exit

DEBUG:
1 parse input
2 set debug status flag if, Y or, N
3 else jump to debug

DUMP:
1 parse filename
2 open file
3 read byte
4 print hex of byte
5 increment till end of file
6 go to 3
7 if EOF close file
8 exit

ERROR:
1 get error number
2 parse range
3 add to error message table
4 print error
5 exit

GO:
1 parse address
2 set return address on stack
3 jump to specified address

HIMEM:
1 parse input
2 if null display current HIMEM status
3 else put new HIMEM value
4 exit

JKL:
1 check if in proper screen mode
2 do screen dump routine
3 exit

KILL:
1 parse filename
2 delete filename
3 exit

LIB:
1 set address of command table
2 display command
3 check for abort
4 increment command pointer
5 if not end of commans go to 2
6 exit

LIST:
1 set the device to CON:
2 parse filename
3 open file
4 read byte
5 display byte to the device
6 increment pointer
7 if not EOF then 4

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BASIC TUTORIAL — Part 4

INPUT, OUTPUT AND DISPLAY

The commands INKEY$, INPUT$, LINE INPUT, LOCATE, OPEN, INPUT# and PRINT# control the input and output of text information for your computer. The formats used here are for cassette based systems. Similar Disk Basic commands will be covered in another section.

The INKEY$ command

The INPUT command discussed in Part 1 allowed the entry of a string of information and would not proceed with the program until the ENTER key was pressed. The INKEY$ command will allow entry of 1 character but will not wait for input before proceeding to the next instruction of the program. Let us compare INPUT with INKEY$.

	INPUT	INKEY$
Displays	?	nothing
Max. characters allowed	256	1
Waits for input	YES	NO
Displays characters entered	YES	NO

Example 1

```
10 INPUT A$
20 PRINT A$
```

Example 2

```
10 A$=INKEY$
20 PRINT A$
```

In the first example, a ? is displayed. If you type in ABC and press ENTER then ABC will be displayed next to the ? and when you press ENTER, then ABC will be displayed again.

Example two works differently. If you are extremely quick, you may be able to enter one character before the program ends - but I doubt it. This is because the INKEY$ in line 10 gives A$ the value of any key which is being pressed at the time that line 10 is performed. Even if you get a key pressed in time, it will only be displayed once, which is done in line 20.

The way to get INKEY$ to wait for input is to place a check in the next line to see if any key has been pressed -

```
10 A$=INKEY$
20 IF A$="" THEN 10
30 PRINT A$
```

This example program will only proceed to line 30 after a key has been pressed. The two sets of quotes (") in line 20 do not have a space between them (NULL STRING), which means IF A$=NOTHING THEN GOTO LINE 10.

The INPUT$ command

This is probably a 'half-way' command between INKEY$ and INPUT. It allows you to specify how many characters will be input. It does not display the characters as they are entered, but it does wait for them to be input. For example -

```
10 A$=INPUT$(3)
20 PRINT A$
```

This program will not proceed to line 20 until three characters have been entered. Nothing will be displayed until line 20 is performed, the same as INKEY$.

A number in brackets must be given and be in the range 1 to 200.

The LINE INPUT command

Also in the explanation of the INPUT command in Part 1, it was mentioned that entering a comma in the input string would result in an 'Extra ignored' message. This is because the comma is used as a 'delimiter' or separator between input variables. For example -

```
10 INPUT A$,B$
20 PRINT A$
30 PRINT B$
```

will display a ? and if you enter ABC,DEF then variable A$ will be ABC and variable B$ will be DEF.

With the LINE INPUT command, you can enter a string which includes commas and they will be accepted as part of the string. For example -

```
10 LINE INPUT A$
20 PRINT A$
```

will, like INKEY$ not display a ? If you enter ABC,DEF then A$ will be ABC,DEF which is what line 20 will print.

The limitations of LINE INPUT are that you can only enter up to 200 characters for the input string and you cannot use it with numeric variables such as LINE INPUT A.

The LOCATE command for Text

We have covered the entry of text, and in each case the print command has displayed the string starting at the left side of the screen. What if we wanted to display something elsewhere. This is where LOCATE command can be used.

The LOCATE command allows you to specify on which line the output will be printed and at what point across the screen. Try this demonstration -

```
10 SCREEN 0
20 WIDTH 40
30 FOR I=1 TO 10
40 LOCATE I,I
50 PRINT I
60 NEXT I
```

LOCATE 0,0 is the top left corner of the screen
LOCATE 0,23 is the bottom left corner of the screen
If the screen WIDTH is set to 40 then
LOCATE 39,0 is the top right corner of the screen
LOCATE 39,23 is the bottom right corner of the screen

The OPEN command

The OPEN command is used to open a channel for use by an external device. It must be used whenever you want to read or write a file on cassette. The term channel, is not quite correct (it is actually the file number used by the computer) but it is used here to avoid confusion with file name.

It takes the form

OPEN "XXXXXX" for Direction as #N

where XXXXXX can be replaced with the name of the file
Direction is either input or output
N is the number of the channel you wish to open

For example - OPEN "NAMES" for output as #1
will open the file NAMES to allow output via channel number 1.

The MAXFILES command

This command allows you to have more than one channel open at a time. It also allocates a section of memory for use as a buffer for each channel. If MAXFILES is not used, then only one buffer area is allocated. An example of using MAXFILES -

```
10 MAXFILES=2
20 OPEN "OLD" FOR INPUT AS #1
30 OPEN "NEW" FOR OUTPUT AS #2
```

If the MAXFILES command is not used, and you specify a channel number greater than 1 then a Bad file number error will result.

The CLOSE command

This command is used to close the open channels and ensure that any information lurking in the buffer area is cleared. It can be used in two ways.

You can close all channels or only specified channels.

```
CLOSE        This will close all open channels
CLOSE #1     This will close channel #1
CLOSE #1,#2  This will close channels #1 and #2
```

If an OPEN command is used for a channel that was already open and has not been closed, then a File already open error will result.

The PRINT# command

To output information to a file that has been opened, we can use the PRINT# command. For example -

```
10 OPEN "FILE1" FOR OUTPUT AS #1
20 PRINT #1,"ABC"
30 PRINT #1,"DEF","GHI"
60 CLOSE
```

This program will send ABCDEFGHI to the file named FILE1. This will be stored on the tape as ABC,DEF,GHI. Notice that a comma has been inserted between each piece of information sent to the cassette. This will be explained in the INPUT# command.

The INPUT# command

This command does the reverse of the PRINT# command. It is used to INPUT information into the computer from a file which has been opened. For example

```
10 OPEN "FILE1" FOR INPUT AS #1
20 INPUT #1,A$,B$,C$
```

If we use the same tape that was used to store the information from the previous example in the PRINT# command, then
string variable A$ will be ABC
string variable B$ will be DEF
string variable C$ will be GHI

so the commas on the tape are ignored. The computer has loaded everything up to, but not including, the comma for each string variable.

Sample program using MAXFILES, OPEN, PRINT#, INPUT# and CLOSE.

```
10 MAXFILES=2
20 A=1:B=2:C=3
30 OPEN "FILE1" FOR OUTPUT AS #1
40 PRINT "PRESS RECORD AND PLAY ON TAPE"
50 PRINT "PRESS ENTER WHEN READY"
60 I$=INKEY$ : IF I$="" THEN 60
70 PRINT #1,A,B,C
80 A=0:B=0:C=0
90 PRINT"REWIND TAPE AND PRESS PLAY"
100 PRINT "PRESS ENTER WHEN READY"
110 I$=INKEY$ : IF I$="" THEN 110
```

```
120 OPEN "FILE1" FOR INPUT AS #2
130 INPUT#1,A,B,C
140 PRINT A
150 PRINT B
160 PRINT C
170 CLOSE
```

COMPUTER MATHS

Don't let the term Computer Maths throw you. In this section we shall be dealing only with the different ways that numbers can be entered into the computer and how it can convert between them. We all had to learn our tables and sums at school without needing any prior knowledge of such things.

Your computer can use four forms of number representation for calculations and other functions. These are the normal Decimal system, the Binary system, the Hexadecimal system and the Octal system.

As you have probably discovered by now, it is possible to use your computer without any knowledge of any other number system except the Decimal system. However, to get the most from your computer and to fully understand how graphics and sprites are handled, it is helpful to at least know about two of the other three. The one we shall skip for now is the Octal system.

DECIMAL, BINARY AND HEXADECIMAL NOTATION

Fortunately, your computer has the built-in ability to convert numbers between any of these forms of notation.

```
Decimal to Binary -      type PRINT BIN$(15)
Binary to Decimal -      type PRINT &B1111
Decimal to Hexadecimal - type PRINT HEX$(15)
Hexadecimal to Decimal - type PRINT &H0F
Binary to Hexadecimal -  type PRINT HEX$(&B1111)
Hexadecimal to Binary -  type PRINT BIN$(&H0F)
```

DECIMAL notation (BASE 10)

The decimal system uses BASE 10 (DECImal) and is written using the numbers 0 to 9. That is, when we reach 9, we move one column to the left -

```
 0
 1          Column No.  3  :  2  :  1  :  0
 .                      ------------------
 .          Decimal No. 1     1     1     1
 9          Value       1000+ 100+  10+   1 = 1111
10
```

As you can see, the number in the second column from the right is the value of 1 times the BASE. This is raised to the power of the number above each column as we move to the left.

10 ^ 2 = 100 or 10 x 10 = 100, 10 ^ 3 = 1000 or 10 x 10 x 10 = 1000

BINARY notation (BASE 2)

The binary system uses BASE 2 (BInary) and is written using the numbers 0 and 1. That is, when we reach 1, we move one column to the left -

```
 0                Column No.   3  :  2  :  1  :  0
 1                             ------------------
10                Binary No.   1     1     1     1
11                Value        8+    4+    2+    1   = 15
100
101               Check this on your computer by typing PRINT &B1111
```

The number in the second column from the right is the value of 1 times the BASE. This is raised to the power of the number above each column as we move to the left.

2 ^ 2 = 4 or 2 x 2 = 4, 2 ^ 3 = 8 or 2 x 2 x 2 = 8

in other words as we move one column to the left, the value DOUBLES.

```
  7   6   5   4   3   2   1   0
---------------------------------
128  64  32  16   8   4   2   1
```

Therefore, placing a 1 in any of these columns means that the value of the amount that would appear in that column must be added to the other values to get the decimal equivalent of the binary number. A zero in any column means do not add the value for that column.

For example -

```
                    7 6 5 4 3 2 1 0                    7 6 5 4 3 2 1 0
                    ---------------                    ---------------
8 bit Binary number 1 1 1 1 1 1 1 1                    1 0 1 0 1 0 1 0
Decimal value =     128+64+32+16+8+4+2+1 = 255         128+0+ 32+0+ 8+0+2+0 = 170
```

To illustrate this further, you may have wondered why some of the commands will only allow a maximum value of 255 and others will allow up to 65535 to be used.

As you may know, the SV & MSX computers are, what is known as, 8 bit computers. That is, they can store 8 'BInary digiTS', or BITS, in each memory location and the largest decimal value that can be held in 8 bits is 255 or 11111111 binary. This can be seen in the example above.

The heart of the computer, the Central Processing Unit or CPU, can address, or 'look at' 16 bit memory locations. The largest decimal value which can be held in 16 bits is 65535.

```
                  15  14  13  12  11  10   9   8   7   6   5   4   3 2 1 0
16 bit           ---------------------------------------------------------
Binary number.    1   1   1   1   1   1   1   1   1   1   1   1   1 1 1 1
```

Decimal = 32768+16384+8192+4096+2048+1024+512+256+128+64+32+16+8+4+2+1=65535

HEXADECIMAL notation (BASE 16)

The hexadecimal (or HEX) system uses BASE 16 (HEX=6, DECI=10) and is written using the numbers 0 to 9 and A to F. That is, when we reach F, we move one column to the left -

```
 0 =  0
 1 =  1
 .    .
 9 =  9
 A = 10
 B = 11          Column No.  3  :  2  :  1  :  0
 C = 12                      ------------------
 D = 13          Hex number  1     1     1     1
 E = 14          Value     4096+  256+  16+   1 = 4369
 F = 15
10 = 16
11 = 17          Check this on your computer by typing PRINT &H1111
 .    .
1E = 30
1F = 31
20 = 32
```

The number in the second column from the right is the value of 1 times the BASE. This is raised to the power of the number above each column as we move to the left.

4	3	2	1	0
65536	4096	256	16	1

GRAPHICS COMMANDS

As mentioned last month, a graphics SCREEN mode must be specified before using the CIRCLE, LINE or DRAW commands.

The CIRCLE command

This takes the form CIRCLE (A,B),C,d,e,f,g
where A is the number of dots across the screen for the centre of the circle
B is the number of dots down the screen for the centre of the circle
C is the number of dots radius for the circle
These three values are compulsory for this command. Optional extras are
d COLOUR (0 to 15)
e START ANGLE 0 to 2*PI (PI=3.14) ie. 0 to 6.28
f END ANGLE 0 to 2*PI (PI=3.14) ie. 0 to 6.28
The start and/or end angles must be given when anything less than a full circle is required.
g ASPECT RATIO (vertical radius/horizontal radius)
The aspect ratio must be given for an ellipse or oval.

The start and end angles can be from 0 to 6.28, where 0 and 6.28 are at the three o'clock position (when compared with a clock), 1.57 is at twelve o'clock, 3.14 is at nine o'clock and 4.69 is at six o'clock. If the start angle is omitted then 0 is assumed and if the end angle is omitted then 6.28 is assumed.

For example -

`CIRCLE (128,96),50`	will draw a circle in the middle of the screen with a radius of 50 dots. The circle will be drawn in the current foreground colour.
`CIRCLE (128,96),50,1`	will draw the same circle in black.
`CIRCLE (128,96),50,1,3.14`	will draw the bottom half of the circle from nine o'clock to the three o'clock position.
`CIRCLE (128,96),50,1,,3.14`	will draw the top half of the circle from three o'clock to nine o'clock.
`CIRCLE (128,96),50,1,1.57,4.71`	will draw the left half of the circle from twelve o'clock to six o'clock.
`CIRCLE (128,96),50,1,,,2`	will draw a vertical oval
`CIRCLE (128,96),50,1,,,1/2`	will draw a horizontal oval. In this example the aspect ratio can be entered as 1/2 or 0.5

This program demonstrates the CIRCLE command.

```
10 SCREEN 1,1 : REM for MSX use SCREEN 2,1
20 COLOR 15,4,4
30 CIRCLE(30,30),10
40 CIRCLE(100,30),10,1
50 CIRCLE(30,60),10,1,3.14
60 CIRCLE(100,60),10,1,1.57
70 CIRCLE(30,90),10,1,4.71
80 CIRCLE(100,90),10,1,3.14
90 CIRCLE(30,120),10,1,,1.57
100 CIRCLE(100,120),10,1,,4.71
105 FOR P=0 TO 6.28 STEP .1 : CIRCLE(200,120),10,1,0,P : NEXT
110 FOR T=1 TO 50 : NEXT
120 FOR A=.6 TO 2 STEP .2 : CIRCLE(A*110,160),10,1,,,A : NEXT
130 GOTO 130
```

The CIRCLE command has a STEP option. This means that the co-ordinates given are taken as being RELATIVE to the last co-ordinates given in a previous CIRCLE, LINE, PSET or DRAW command. For example -

```
10 SCREEN 1,1 : REM MSX SCREEN 2,1
20 CIRCLE (20,20),10,15
30 GOSUB 110
40 LINE (0,100)-(50,100),15
50 GOSUB 110
60 PSET (100,50),15
70 GOSUB 110
80 DRAW "BM150,100C15D50"
90 GOSUB 110
100 GOTO 100
110 CIRCLE STEP(10,10),10,1
120 RETURN
```

Line 110 in this example uses the STEP option to draw a circle with its centre 10 dots down and 10 dots to the right of the co-ordinates given in the command preceding the GOSUB.

SOUND COMMANDS

More on the SOUND command

Registers 6 and 7 control the noise generator and its mixing with the tones produced on the three channels.

REGISTER 6

This register controls the frequency of the noise generator and can hold a value between 0 and 31.

```
10 SOUND 0,100
20 SOUND 1,10
30 SOUND 7,0
40 SOUND 8,8
50 SOUND FOR F=0 TO 31 STEP 4
60 SOUND 6,F
70 A$=INPUT$(1)
80 NEXT F
```

This program will produce a fluttering sound that changes slightly when any key is pressed.

REGISTER 7

This is the mixer register and controls the mixture of tone and noise on each of the three channels. The value of the byte contained in this register is "bit oriented", that is each bit of the byte serves a purpose. As you may now be aware, a byte contains eight bits with each bit having a doubling value from the right, and a byte can hold a maximum value of 255 -

BIT NUMBER	8	7	6	5	4	3	2	1
BINARY NUMBER	1	1	1	1	1	1	1	1
VALUE if 1 in bit	128	+64	+32	+16	+8	+4	+2	+1 = 255

TONE ON/OFF	BIT 1 - A zero means tone on Channel A BIT 2 - A zero means tone on Channel B BIT 3 - A zero means tone on Channel C
NOISE ON/OFF	BIT 4 - A zero means noise on Channel A BIT 5 - A zero means noise on Channel B BIT 6 - A zero means noise on Channel C
	BIT 7 - zero or one have no effect BIT 8 - zero or one have no effect

Example -

```
10 SOUND 0,100
20 SOUND 1,10
30 SOUND 6,20
40 SOUND 8,8
50 SOUND 7,55 : REM BINARY 110111
60 PRINT "NOISE ON CHANNEL A"
70 GOSUB 150
80 SOUND 7,62 : REM BINARY 111110
```

```
90 PRINT "TONE ON CHANNEL A"
100 GOSUB 150
110 SOUND 7,54 : REM BINARY 110110
120 PRINT "TONE & NOISE ON CHANNEL A"
130 GOSUB 150
140 END
150 FOR T=1 TO 1000 : NEXT T
160 RETURN
```

The PLAY command

This command accesses the MUSIC LANGUAGE of the computer. It takes the form -

PLAY X---X,Y---Y,Z---Z

X---X is a string of command letters, numbers and symbols for channel 1.
Y---Y is a string of command letters, numbers and symbols for channel 2.
Z---Z is a string of command letters, numbers and symbols for channel 3.

For example PLAY "C","D","E"

Plays the note C on channel 1, D on channel 2 and E on channel 3

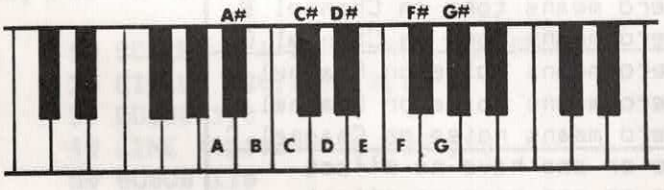
A missing value for a channel will cause a Syntax error.

For example - PLAY "C",,"E"

The information to be supplied to each channel consists of -

A to G The musical notes to be played. These are in the range A to G and use the # or + signs to indicate Sharps and the - sign to indicate Flats. For example -

A A# B C C# D D# E F F# G G#



O number The Octave number for the notes that follow. This can be in the range 1 to 8. If no Octave is specified at the start, then the previous octave setting will be used. If this is the first time the PLAY command is encountered then Octave 4 is assumed. For example -

PLAY "AB02AB06AB"

will play AB in octave 4, AB in octave 2 and AB in octave 6

Note: O4C is middle C on a piano.

N number This will play the note given by the number from the table below. For example N55 will play the D note from octave 4.

01	02	03	04
CC#DD#EFF#GG#AA#B	CC#DD#EFF#GG#AA#B	CC#DD#EFF#GG#AA#B	CC#DD#EFF#GG#AA#B
11111111	1122222222233333	33333444444444455	55555555666666666
1234567890123456	7890123456789012	3456789012345678	9012345678901234

05	06	07	08
CC#DD#EFF#GG#AA#B	CC#DD#EFF#GG#AA#B	CC#DD#EFF#GG#AA#B	CC#DD#EFF#GG#AA#B
	111	1111111111111111	1111111111111111
6777777777888888	8888999999999000	0000000111111111	2222222223333333
9012345678901234	5678901234567890	1234567890123456	7890123456789012

Note: N52 is middle C on a piano.

L number This specifies the LENGTH of the note and number can be 1 to 64. If no L value is given then 4 or quarter note is assumed.

L1 = whole note Semibreve
L2 = half note Minim
L4 = quarter note Crotchet
L8 = one eighth note Quaver
and so on...

The length value can also be given after the note, A4 is the same as using L4A

R number The REST command. The number is in the range of 1 to 64 and is calculated the same as for the LENGTH command. If no number is given then 4 is assumed.

T number This is the TEMPO command which sets the number of quarter notes per minute. The number can be 32 to 255. If no Tempo command is given, then T120 is assumed.

V number VOLUME command. Sets the output volume and can have a number from 1 to 15. If no volume command is given, then V8 is assumed.

M number MODULATION command (also known as tremolo). The number can be from 1 to 65535. If no number is given, then 255 is assumed.

S number This sets the SHAPE for the envelope. The number used can be from 1 to 15. For the effect of each number, see the SOUND command in last months' tutorial pages 20 and 21.

If the same piece of music is to be played more than once, then it can be defined as a string variable for use with the PLAY command. For example -

```
100 A$="AA#BCC#DD#EFF#GG#"
110 PLAY A$
120 PLAY A$
```

QUESTIONS

1. Will this display a ? while waiting for input?

```
10 INPUT X
20 PRINT X
```

2. Will this wait for input?

```
10 I$=INKEY$
20 PRINT I$
```

3. How many values can be entered in this?

```
10 INPUT A,B,C
20 PRINT A,B,C
```

4. Where will COMPUTER FORUM be displayed on the screen?

```
10 SCREEN 0
20 LOCATE 13,12
30 PRINT "COMPUTER FORUM"
```

5. What is wrong with this?

```
10 OPEN "ALPHA" FOR OUTPUT AS #1
20 OPEN "BETA" FOR OUTPUT AS #1
```

6. Give the decimal values of ... (a) &B110011 (b) &H0A

7. Give the binary values of ... (a) 10 decimal (b) 14 decimal

8. Give the hexadecimal values of ... (a) 12 decimal (b) 15 decimal

9. Will this draw the top or bottom half of a circle?

```
CIRCLE (128,96),50,1,3.14
```

10. If, in a series of sound commands, SOUND 7,62 is used, which channel will have tone but no noise?

8 close file
9 exit

LOAD: 1 parse filename
2 read file
3 setup run addresses
4 exit

THE SV ROM DECODED

Here is a section of the SV ROM showing a brief outlay of the routines and data areas.

```
          SPECTRAVIDEO ROM VERSION 1.1

              GENERAL ROM LAYOUT

0000  Boot vector          0907  ERROR HANDLER entry
0003  CINT address         09A0  print error
0005  PUTHL address        09AF  COMMAND LINE
0008  RST 08H              0B27  check if line exists
0010  RST 10H              0B44
0018  RST 18H              0CF5  Base converter
0020  RST 20H              0D1F  ELSE entry 2
0026  LINK START address   0D2C
0028  RST 28H              0D65  FOR
0030  RST 30H              0F5C  DEFSTR entry
0038  RST 38H  INTERRUPT   0F5F  DEFINT entry
            entry          0F62  DEFSNG entry
003B  BIOS table           0F65  DEFDBL entry
0066  NMI vector           0FE2  RUN
0069  BIOS table contd      0FF6  GOSUB
0180  NMI routine          1028  GOTO
0185  Command  addresses   1061  RETURN
0235  Function addresses   109B  DATA
0295  Code     addresses   10C0  LET
02C9  Command strings      1124  ON
          and tokens       114D  ON continued
0587  Data tables          119D  RESUME
05D6  Error messages       11EA  ERROR
084C  Data table           11F5  AUTO
089A  'in'     message     1225  IF
089F  'Ok'     message     125D  LPRINT
08A4  'Break'  message     1265  PRINT
08AA  Routine              12DE  ',' handler for PRINT
08C9                       132C  TAB handler for PRINT
08ED                       1360  ';' handler for PRINT

      ***** The end of instalment 1 *****
```

Also the rest of the Spectravideo keywords and their corresponding addresses —

KEYWORD	TOKEN	ADDRESS	STR$	255,147	6909
SGN	255,132	55C6	STRIG	255,163	3263
SIN	255,137	50D1	SWAP	164	6735
SOUND	196	2BFD	TAN	255,141	5120
SPACE$	255,153	6B4D	USR	221	1842
SQR	255,135	5222	VAL	255,148	68C0
STICK	255,162	3206	VPEEK	255,152	46F2
STOP	144	66C8	VPOKE	198	46D8

THE VIDEO DISPLAY PROCESSOR

(courtesy of Larry Dunning)

Inside your computer controlling all the video involved in making fantastic games surpassing the CBM64 is the VDP or Video Display Processor.

A general description of its functions.

Four screen modes:

Text	40 by	24 characters
Graphics I	32 by	24 characters
Graphics II	256 by	192 pixels
Multicolor	64 by	48 pixels

32 sprites with sizes of 8 by 8 or 32 by 32 pixels and magnification.
Eight internal control registers.
16K of graphics memory is used to keep the image.
External video input (not supported in SV or MSX1)

THE CONTROL REGISTERS

Note: They are write only.

Reg.	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	M3	EV
1	K	BL	IE	M1	M2	0	SZ	MG
2	0	0	0	0	N	N	N	N
3	C	C	C	C	C	C	C	C
4	0	0	0	0	0	P	P	P
5	0	S	S	S	S	S	S	S
6	0	0	0	0	0	G	G	G
7	F	F	F	F	B	B	B	B
VDP	FL	5S	C	X	X	X	X	X

Note: The VDP register is read only.

M3, M2 & M1 determine the screen mode.

0	0	0	= Graphics I
1	0	0	= Graphics II
0	1	0	= Multicolor
0	0	1	= Text

Register 0 — Control register 1

EV = 1, external video display active
EV = 0, external video display inactive

Register 1 — Control register 2

BIT 7 — K = 1, 16K maximum memory size
K = 0, 4K maximum memory size
BIT 6 — BL = 1, enables the display
BL = 0, disables the display
BIT 5 — IE = 1, video interrupts enabled
IE = 0, video interrupts disabled
BIT 1 — SZ = 1, 32 x 32 size sprites
SZ = 0, 8 x 8 size sprites
BIT 0— MG = 1, double size sprites
MG = 0, normal size sprites

Register 2 — Name Table (NTB)

NNNN defines the base address for the NAME TABLE (NTB) multiply this value by 400H to find its starting address in video memory.

Register 3 — Color Table (COLB)

CCCCCCCC defines the base address for the COLOR TABLE (COLB) multiply this value by 40H to find its starting address in video memory.

Register 4 — Pattern Generator Table

PPP defines the base address for the PATTERN GENERATOR TABLE (PGB) multiply this value by 800H to find its starting address in video memory.

Register 5 — Sprite Attribute Table

SSSSSSS defines the base address for the SPRITE ATTRIBUTE TABLE (SAB) multiply this value by 80H to find its starting address in video memory.

Register 6 — Sprite Pattern Generator Table (SPGB)
GGG defines the base address for the SPRITE PATTERN GENERATOR TABLE (SPGB) multiply this value by 800H to find its starting address in video memory.

Register 7 — The COLOR register
FFFF defines the foreground color while in text mode. BBBB defines the background color in text mode, or the border color in other modes.

The VDP Register
FL = end of raster flag
C = two sprite collision flag
5S = flag set when 5 or more sprites are on one line
XXXXX = number of the sprite causing problems when the 5S flag is on

ACCESSING THE REGISTERS
MSX — Information stored in the Write-only registers, 0-7 can be altered using the VDP command. For example, to change register 1, use VDP (1) = 224 (&B11100000). To examine the information stored in the Read-only VDP register 8, use PRINT VDP (8). The effects of altering registers will be shown in further discussions of the VDP chip.

SPECTRAVIDEO — Unfortunately, the SV has no BASIC command for accessing the VDP registers. A method for doing so will be included in next month's column.

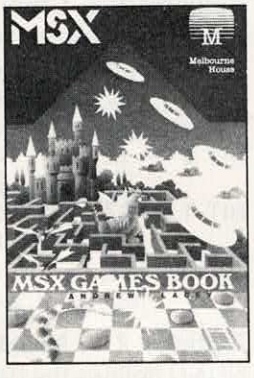
MSX SCREEN DISABLE
Type in and RUN this program to observe the effects of the screen disable and enable ROM calls at &h41 and &h44.

```
10 SCREEN 2,1
20 DEFUSR=&H41:J=USR(0)
30 CIRCLE (128,96),50,1
40 PAINT (128,96),1
50 DEFUSR=&H44:J=USR(0)
60 GOTO 60
```

Well finished another issue — Whew!! Until next time hackers and beginners — ADIOS!

BOOKS • BOOKS • BOOKS • BOOKS

MSX GAMES BOOK
by Andrew Lacey

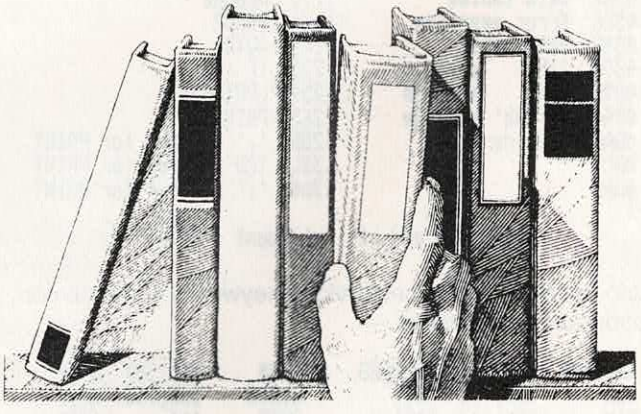


This book contains 27 games programs of several types ranging from interactive graphics through to educational and shoot-out games. Here are the titles —

Patterns, Astral Attack, Cosmic Critters, Planetoid, Balloons, Soccer Pro, Sea Harrier, Secret Ship-Shuttle Search, Shooting Gallery, UFO, Robot Raiders, Laser Battle, Dragster, Mind Quiz, Concentration, Stream Hopper, Chopper Landing, Tank Ambush, Stunt Man, Cross Country, Alligator, Convoy, Gorilla Maths, Junior Minotaur Mastermind, Minotaur Mastermind, Treasure Trek, Thirty-One.

Two entry programs are included at the start of the book. The first of these, FINDER, checks your computer to find where the start of a basic program is in memory. The second is a checksum program which can be used to see if you have typed in the programs correctly. This is a handy feature.

We chose three games to type in — Convoy, Sea Harrier and Chopper Landing. None of these worked on our computer UNTIL we disabled the disk system. The machine code support program loads into memory where the disk operating system RAM resides. To use this book with a disk system would require the support program to be moved and each program that calls it would need modifying.



Apart from that, we found the games worth the effort of typing in. All except seven of these games require the use of the machine code support program which is contained in one of the appendices. Therefore we do not know, at this stage, if these programs are suitable for conversion to Spectravideo format.

Review copy supplied by Peter J. Fox Computers. Retail price $17.25.

Publisher: Melbourne House
224 pages

HINTS & TIPS

SCREEN DUMP

by Rod Johnson

On reading Mr Harry Tarasinski's letter in Vol. 1, No. 2 edition (re screen print for graphics tablet) I was prompted to write a screen dump program in BASIC.

On completion of the program I found that the printout had a few errors. After investigation I found that the Spectravideo BASIC interpreter has an error. Not a "bug" as such but an error in the programming (Microsoft BASIC on the IBM does not have the same error).

When you PRINT chr$(9); "X" the cursor is tab'd eight columns before the X is printed, this is quite correct as chr$(9) is the TAB character. When you LPRINT chr$(9); "X" the print head is tab'd eight columns before the "X" is printed, this is not correct as the printer has its own tab command and the computer should not output eight spaces for the tab function. This is not a problem you are probably thinking. Well if you are doing a graphics dump to the printer it expects characters AS THEY ARE — not with an extra eight chr$(32) every time it should be chr$(9).

To explain. If the computer sends an ASCII 1 to the printer, on most printers, this will operate the bottom pin of the print head and ASCII 255 will operate all the pins. The relationship can be seen when the characters are represented in their binary form, 00000001 (1) operates the bottom pin and 11111111 (255) operates all eight pins. When the character 00001001 (9) is sent you expect the bottom and fourth pins to operate but with the SV318/328 BASIC eight spaces are sent instead (00100000 (32)). This produces a line eight dots wide or every character 9 that should have been printed. As the printer only expects a set number of characters, per graphics line, the print out becomes a mixture of wrong graphics and ASCII characters.

And so to the program and line 130 is the way I overcame the problem. I changed every character 9 to 13, this produces a thick line above the bottom line but as a compromise it isn't bad.

```
10 DEFINT A-Z:B=2
20 SCREEN 1 : REM MSX SCREEN 2
30 GOSUB 200
40 LPRINT CHR$(27)+"3"+CHR$(24);
50 FOR Z=0 TO 191 STEP 8
60 LPRINT CHR$(27);"*";CHR$(5);CHR$(0);C
HR$(1);
70 FOR X=0 TO 255
80 D=0:P=8
90 FOR Y=Z TO Z+7
100 A=POINT (X,Y):P=P-1
110 IF A=4 THEN A=0 ELSE A=B^P
120 D=D+A:NEXT Y
130 IF D=9 THEN D=13
140 LPRINT CHR$(D);:NEXT X
150 LPRINT CHR$(10);:NEXT Z
160 LPRINT CHR$(27);"2";CHR$(12)
170 END
180 REM
190 '******** Test Program ********
200 LOCATE 96,6:PRINT "Test Pattern"
210 CIRCLE (128,96),20,15
220 CIRCLE (128,96),40,15
230 PAINT (150,110),15
240 CIRCLE (128,96),60,15
250 CIRCLE (128,96),80,15
260 CIRCLE (128,96),100,15
270 CIRCLE (128,96),120,15
280 LINE (0,0)-(255,191),15
290 LINE (0,0)-(0,191),15
300 LINE (0,0)-(255,0),15
310 LINE (255,191)-(255,0),15
320 LINE (0,191)-(255,191),15
330 LINE (255,0)-(0,191),15
340 LINE (78,46)-(178,146),15,B
350 LINE (0,96)-(256,96),15
360 LINE (128,46)-(128,191),15
370 RETURN
```

PROGRAM EXPLANATION

I will explain how the program works for anyone who wants to try it on a different printer. First the program runs EPSON printers and many others that have the same control codes. It also takes about thirty seconds to print a line of graphics (due to the large number of calculations per line — 256 dots x 8 = 2048 per line), that's about ten minutes per screen dump.

Line 40 sets the spacing between lines to the height of the print head (24/216 inch). Line 60 sets the printer graphics mode with a dot density of 576 dots per line or 72 dots per inch, with the number of dots per line set at 256. This is sent to the printer at the start of every line of graphics. Line 110 produces a number equivalent to the binary position of the pin on the print head. If the colour on the screen is the background colour (in the program it is colour 4) it produces a 0 and no pin is operated.

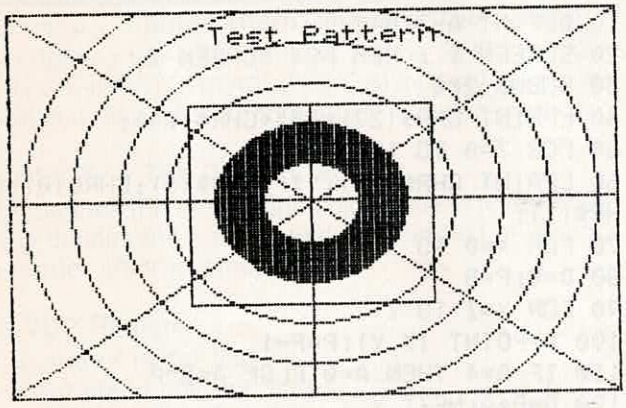


Diagram 1

Line 120 totals the numbers to produce a character to be sent to the printer equivalent to the binary representation of a column of eight dots on the screen.
Line 130 replaces the character 9 with character 13.
Line 140 sends the character to the printer.
Line 150 produces a line feed at the end of the graphics line.
Line 160 resets the line spacing to 1/6 inch and produces a form feed.

Variable X is the horizontal position on the screen.
Variable Z is the top dot of the column being calculated.
Variable Y is the vertical position on the screen.

The screen is interrogated in columns of eight dots and a character is produced for every column. At the end of the line the Z variable is incremented by eight and the X variable is reset to zero. This produces a zig-zag scanning pattern which sends 256 characters to the printer for every line.

If the pattern produced looks like diagram 2 then change line 80 to-80 D=0: P= −1
and line 100 to — 100 A = POINT (X,Y): P=P+1.

This is because the printer you are using expects a character 1 to fire the top pin of the print head instead of the bottom pin. Hope this is a help for Mr Tarasinski.

Rod Johnson
Blackburn North, Victoria

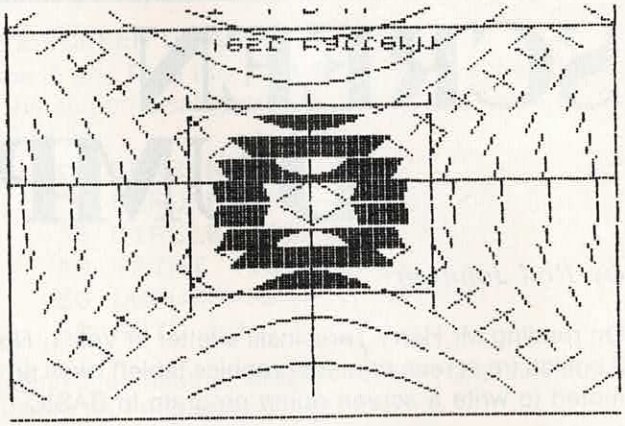


Diagram 2

Thank you for the explanation and screen dump program Rod. I am sure it will be of use to Harry and quite a few of our readers as it is not limited to use with the graphics tablet and can be used as a screen dump program under other circumstances.

As Rod points out, you may need to alter some of the LPRINT statements for it to work properly with your particular printer. To get the program to operate properly with our STAR DP515 printer we had to make the following changes —

40 LPRINT CHR$(27) + "3" + CHR$(16);
60 LPRINT CHR$(27); "K"; CHR$(0); CHR$(1);

USER GROUPS

Brisbane Spectravideo & MSX Users Group
C/- Mrs Lucille Parker (Secretary)
25 Primrose Street
Woodridge
Queensland 4114

Melbourne Spectravideo & MSX Users Group C/- John Halford, 48 Shawlands Ave, Blackburn South, Victoria 3130. Phone (03) 878 6169. Meeting on the first Saturday afternoon of each month at the Balwyn Community Centre, Whitehorse Road, Balwyn, sharing public domain software and looking at new equipment.

The MSX Users Group
C/- Max Cowin
P.O. Box 87
Mt Hawthorn
W.A. 6016

Wellington Spectravideo Users Group
P.O. Box 26-050
Newlands
Wellington 6004
New Zealand

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COMPUTER KIDS

Hi kids! It's Wendy with you again in issue four. Well it's been a bit of a cold, wet month, but most of us had our school holidays, so at least we could be having a good time while we got rained on.

This month you get another 'Wordsearch' puzzle, and also a short program that shows you the colours that your computer can use. How did you go on last month's 'Wordsearch'? I made 18 words. I bet you did better than that!

I'll join you next issue! Bye for now. Wendy.

WORDSEARCH

Your word is: **HEXADECIMAL**

```
10 COLOR 15,15,15
20 SCREEN 3,2:REM FOR SV SCREEN 2,2
30 FOR I=1 TO 8
40 READ A
50 A$=A$+CHR$(A)
60 NEXT I
70 SPRITE$(0)=A$
80 FOR C=1 TO 14
90 FOR P=1 TO 200 STEP 4
100 PUT SPRITEC,(P+4,C*8),C,0
110 LINE (0,C*8+2)-(P,C*8+6),C,BF
120 NEXT P
130 NEXT C
140 GOTO 140
150 DATA 4,14,31,62,124,248,240,224
```

SOUND CENTRE

Sound routines submitted by our readers.

SOUND CENTRE

Paul Rode of Curlewis has sent in these routines.

Banjo.

```
10 PLAY"S10M5CDECCDEC","S1M5000CDECCDEC"
```

Bagpipes.

```
10 PLAY"S12M5000CDECCDEC","S11M500CDECCD
EC"
```

```
10 REM OCEAN SOUND
20 SOUND 0,0
30 SOUND 6,10
40 SOUND 7,16
50 SOUND 8,16
60 SOUND 12,85
70 SOUND 13,14
```

```
10 REM SHOTGUN
20 SOUND 0,0
30 SOUND 6,10
40 SOUND 7,20
50 SOUND 8,16
60 SOUND 12,30
70 SOUND 13,0
```

For a machine gun, use the shotgun sound repeated about ten times with a delay between each shot.

```
10 REM PASSING PLANE
20 SOUND 5,6
30 SOUND 6,12
40 SOUND 7,2
50 SOUND 10,16
60 SOUND 12,203
70 SOUND 13,14
```

```
10 REM HELICOPTER
20 SOUND 6,37
30 SOUND 7,35
40 SOUND 8,19
50 SOUND 12,4
60 SOUND 13,12
```

For the Helicopter sound, try changing SOUND 12,4 to SOUND 12,8. This will have the effect of changing the speed.

Written any programs? Send them in to COMPUTER FORUM for inclusion in our Program Listings. If we decide to publish, you will receive payment based on size, originality and scope of interest. Please send programs in cassette form or disk and clearly marked which format they are for. If you wish them to be returned, enclose sufficient postage for safe return. Mark your name and address on everything.

CLUB NEWS

MSX USERS GROUP (W.A.)

The MSX Users Group welcomes Spectravideo 318/328 owners!

At the June meeting (our second at the new venue) it was unanimously decided to expand membership to include early SVI owners, in line with several other groups around Australia. As this magazine itself bears out, the advantages are enormous in a sharing of BASIC software and ideas.

Also decided at the meeting was to print membership cards, as we have several retailers willing to offer a club discount to cardholders. This will also ensure that only financial members will be able to take advantage of club benefits.

The highlight of the meeting was the arrival of Terry Brown, Sales Manager of Sanyo in W.A. As well as showing a personal interest in learning about MSX as a user, he has offered the club the use of some of the new Sanyo MSX range at meetings, and had a few thoughts about possible club projects. Also present to lend his support was our illustrious editor, Brod Brown.

Don't forget folks — meetings are the second TUESDAY of every month, at the LEEDERVILLE TOWN HALL (Lesser Hall), 84 Cambridge Street, Leederville. If you haven't joined yet, come to the next meeting anyway, and see for yourself how much fun it is.

'Til next time — Erik Kowarski.

CANADIAN SV & MSX USERS GROUP

Recently received in this office was a copy of a newsletter from SMUG DE MONTREAL. This was written half in English and half in French (fortunately the front half is English and the other half is French). They appear to be a fairly active group having connections with MSX groups in Holland, Belgium and Denmark.

It would appear that Canadians too have their fair share of problems obtaining software and information for SVI and MSX. If you would like to correspond with them, their address is:

SMUG de Montreal
443 Boulevard Roche
Vaudreuil (Quebec)
Canada J7V 2N4

READERS ADS

Use these pages to sell your surplus equipment, etc. Photocopy this form and insert up to 20 words. Put it in an envelope with $5 and send to COMPUTER FORUM by the 10th of each month.

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SYNTAX ERRORS etc.

It appears that our program GOPHER (Vol. 1 No. 2) had a couple of bugs in it for MSX owners.

Line 400 — LOCATE 210,I should be changed to PRESET (210,I) and PRINT M$ should be PRINT#1,M$

MSX alterations line 570 change AS#1,: to AS#1 :

Also line 605 to be inserted as 605 KEYOFF

Our thanks to Ian Davidson, of Mitchum, Victoria for bringing them to our attention.

PROGRAM LISTINGS

Each issue of COMPUTER FORUM will contain program listings for you to type in. When entering these programs it is a good idea to save off to cassette or disk every 10 or 20 lines and then continue. Then if the power goes off or another major disaster occurs, you will not have to type in the whole lot again.

Another check is to verify each save using the CSAVE? command. To use this command, CSAVE your program, then rewind the tape and type CLOAD? and press enter. This command will then compare the program on the tape with the one in memory, to verify that it was saved correctly.

All of the listings are printed in 40 column format. If you enter WIDTH40 before typing them in, you will be able to line up the characters in the listing with the text on screen.

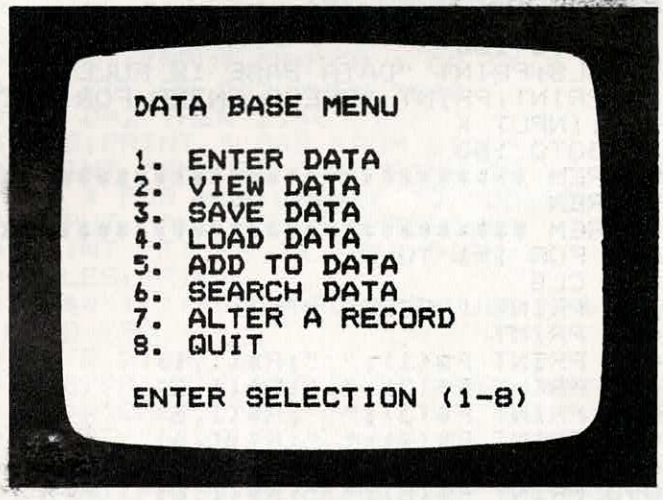
Take care that you do not mistake the letter l for 1 or an O for a zero. Where graphics characters are used in the program, they will be shown as @1@. This symbol will then be repeated at the end of the listing with instructions concerning the graphics character used.

All of the listings have been tested on MSX and SV318/328 computers and do work. Should you have any problems, we suggest that you thoroughly check your work as errors can creep in very easily. Should you not wish to type in these listings, send a blank tape together with $2 to COMPUTER FORUM and we will post a copy to you. Do not forget to tell us which computer you have.

DATABASE

THIS IS A DATA ENTRY AND RETRIEVAL PROGRAM WHICH ALLOWS YOU TO ENTER, SAVE, LOAD, SEARCH AND ALTER RECORDS HAVING UP TO SIX ENTRY FIELDS. THIS IS NOT A COMPLETE DATA BASE AS IT DOES NOT HAVE SORT OR PRINTING ROUTINES.
THE PURPOSE OF PUBLISHING THIS PROGRAM IS TO SHOW THE MINIMUM REQUIREMENTS OF A DATA BASE PROGRAM AND TO STIMULATE YOUR INTEREST IN MODIFYING PROGRAMS FOR YOUR OWN USE.

This program is suitable for use on SV318, SV328 and MSX.




```
100 CLEAR 6000:DEFINT A-Z:WIDTH 39:DIM F
$(6),R$(50,6)
110 FOR I=1 TO 6:READ F$(I):NEXT
120 REM CHANGE THE SIX DATA ITEMS IN
130 REM LINE 150 TO WHATEVER FIELD
140 REM NAMES YOU REQUIRE
150 DATA NAME,ENGLISH,MATHS,SCIENCE,SOCI
AL STUDIES,HISTORY
```

```
160 CLS
170 PRINT TAB(10);"DATA BASE MENU"
180 PRINT
190 PRINT TAB(10);"1. ENTER DATA"
200 PRINT TAB(10);"2. VIEW DATA"
210 PRINT TAB(10);"3. SAVE DATA"
220 PRINT TAB(10);"4. LOAD DATA"
230 PRINT TAB(10);"5. ADD TO DATA"
240 PRINT TAB(10);"6. SEARCH DATA"
250 PRINT TAB(10);"7. ALTER A RECORD"
260 PRINT TAB(10);"8. QUIT"
270 PRINT
280 PRINT TAB(10);:INPUT"ENTER SELECTION
 (1-8)";S
290 IF S<1 OR S>8 THEN 160
300 ON S GOTO 500,1000,1500,2000,510,250
0,3000,3500
490 REM ********************************
491 REM          INPUT DATA
492 REM ********************************
500 R=0
510 IF R=40 THEN 650
520 CLS
530 PRINT "TO END, ENTER 0 FOR NAME":PRI
NT
540 PRINT F$(1);:INPUT N$:IF N$="0" THEN
 GOTO 640 ELSE R=R+1:R$(R,1)=N$
550 REM IF NAME=0 THEN GO TO MENU
560 PRINT F$(2);:INPUT R$(R,2)
570 PRINT F$(3);:INPUT R$(R,3)
580 PRINT F$(4);:INPUT R$(R,4)
590 PRINT F$(5);:INPUT R$(R,5)
600 PRINT F$(6);:INPUT R$(R,6)
610 PRINT "PRESS ENTER FOR NEXT RECORD";
:INPUT X
620 IF R<40 THEN 520
630 Z=FRE(X$)
640 GOTO 160
650 CLS:PRINT "DATA BASE IS FULL!"
660 PRINT:PRINT "PRESS ENTER FOR MAIN ME
NU";:INPUT X
670 GOTO 160
990 REM ********************************
991 REM          VIEW DATA
992 REM ********************************
1000 FOR I=1 TO R
1010 CLS
1020 PRINT "RECORD #";I
1030 PRINT
1040 PRINT F$(1);" ";R$(I,1)
1050 PRINT F$(2);" ";R$(I,2)
1060 PRINT F$(3);" ";R$(I,3)
1070 PRINT F$(4);" ";R$(I,4)
1080 PRINT F$(5);" ";R$(I,5)
1090 PRINT F$(6);" ";R$(I,6)
1100 PRINT
1110 PRINT TAB(10);:INPUT "PRESS ENTER F
OR NEXT RECORD";X
1120 NEXT
1130 PRINT:PRINT "END OF DATA"
1140 PRINT:PRINT "PRESS ENTER FOR MAIN M
ENU";:INPUT X
1150 GOTO 160
```

```
1490 REM ******************************
1491 REM            SAVE DATA
1492 REM ******************************
1500 CLS:PRINT "SAVE DATA"
1510 D=0:NM$="":PRINT
1520 PRINT "1. SAVE TO DISK"
1530 PRINT "2. SAVE TO TAPE"
1540 PRINT:INPUT "ENTER 1 OR 2";D
1550 IF D<1 OR D>2 THEN 1500
1560 IF D=2 THEN 1640
1570 CLS:PRINT "SAVE TO DISK"
1580 PRINT "ENSURE DATA DISK IS IN DRIVE
 1":REM * FOR MSX CHANGE TO "DRIVE A"
1590 INPUT "AND PRESS ENTER";X
1600 PRINT "FILES ON THIS DISK..."
1610 FILES:PRINT
1620 NM$="1:":REM * FOR MSX USE NM$="A:"
1630 GOTO 1680
1640 CLS:PRINT "SAVE TO TAPE":MOTOROFF
1650 NM$="CAS:"
1660 PRINT "GIVE FILE NAME FOR THIS DATA
"
1670 INPUT FL$
1680 PRINT:PRINT "ENSURE TAPE IS REWOUND
 AND PRESS"
1690 PRINT "RECORD AND PLAY"
1700 PRINT:INPUT "PRESS ENTER";X
1710 OPEN NM$+FL$ FOR OUTPUT AS #1
1720 PRINT #1,R
1730 FOR I=1 TO R:FOR J=1 TO 6
1740 PRINT #1,R$(I,J)
1750 NEXT J,I
1760 CLOSE #1
1770 GOTO 160
1990 REM ******************************
1991 REM            LOAD DATA
1992 REM ******************************
2000 CLS:PRINT "LOAD DATA"
2010 D=0:NM$="":PRINT
2020 PRINT "1. LOAD FROM DISK"
2030 PRINT "2. LOAD FROM TAPE"
2040 PRINT:INPUT "ENTER 1 OR 2";D
2050 IF D<1 OR D>2 THEN 2000
2060 IF D=2 THEN 2140
2070 CLS:PRINT "LOAD FROM DISK"
2080 PRINT "ENSURE DATA DISK IS IN DRIVE
 1":REM * FOR MSX CHANGE TO "DRIVE A"
2090 INPUT "AND PRESS ENTER";X
2100 PRINT "FILES ON THIS DISK..."
2110 FILES:PRINT
2120 NM$="1:":REM * FOR MSX USE NM$="A:"
740 GOTO 790
2130 GOTO 2160
2140 CLS:PRINT "LOAD FROM TAPE":MOTOROFF
2150 NM$="CAS:"
2160 PRINT "GIVE FILE NAME TO LOAD"
2170 INPUT FL$
2180 PRINT:PRINT "ENSURE TAPE IS REWOUND
 AND PRESS"
2190 PRINT "PLAY"
2200 PRINT:INPUT "PRESS ENTER";X
2210 OPEN NM$+FL$ FOR INPUT AS #1
2220 INPUT #1,R
2230 FOR I=1 TO R:FOR J=1 TO 6
2240 INPUT #1,R$(I,J)
2250 NEXT J,I
2260 CLOSE #1
2270 GOTO 160
```

```
2490 REM ******************************
2491 REM          SEARCH DATA
2492 REM ******************************
2500 CLS
2510 PRINT "SEARCH ROUTINE"
2520 PRINT:PRINT "SEARCH ON WHICH FIELD"
2530 FOR I=1 TO 6
2540 PRINT I;". ";F$(I)
2550 NEXT
2560 PRINT:INPUT "SELECT 1-6";F
2570 IF F<1 OR F>6 THEN 2500
2580 CLS:PRINT "ENTER DATA FOR SEARCH"
2590 PRINT "THE INFORMATION MUST BE EXAC
TLY THE"
2600 PRINT "SAME AS THE RECORD. IF YOU E
NTER UPPER"
2610 PRINT "CASE AND THE RECORD IS STORE
D IN lower"
2620 PRINT "case THEN IT WILL NOT BE FOU
ND!"
2630 PRINT:INPUT "ENTER INFORMATION";S$
2640 FR=0:FOR I=1 TO R
2650 IF R$(I,F)=S$ THEN FR=I:I=R
2660 NEXT
2670 IF FR=0 THEN PRINT "RECORD NOT FOUN
D!":GOTO 2710
2680 CLS:PRINT "RECORD #";FR
2690 PRINT:FOR I=1 TO 6
2700 PRINT F$(I);" ";R$(FR,I)
2710 NEXT
2720 PRINT:PRINT "PRESS ENTER FOR MAIN M
ENU";:INPUT X
2730 GOTO 160
2990 REM ******************************
2991 REM          ALTER A RECORD
2992 REM ******************************
3000 CLS:PRINT "ALTER A RECORD"
3010 PRINT
3020 INPUT "ENTER NUMBER OF RECORD";RN
3030 PRINT "CURRENT DETAILS ARE -"
3040 FOR I=1 TO 6
3050 PRINT F$(I);" ";R$(RN,I)
3060 NEXT
3070 PRINT
3080 PRINT "ENTER NEW DETAILS FOR EACH F
IELD"
3090 PRINT "OR PRESS ENTER TO LEAVE UNCH
ANGED."
3100 PRINT F$(1);:INPUT R$(RN,1)
3110 PRINT F$(2);:INPUT R$(RN,2)
3120 PRINT F$(3);:INPUT R$(RN,3)
3130 PRINT F$(4);:INPUT R$(RN,4)
3140 PRINT F$(5);:INPUT R$(RN,5)
3150 PRINT F$(6);:INPUT R$(RN,6)
3160 PRINT:PRINT "PRESS ENTER FOR MAIN M
ENU";:INPUT X
3170 GOTO 160
3490 REM ******************************
3491 REM                QUIT
3492 REM ******************************
3500 CLS:PRINT "BYE FOR NOW!"
3510 END
```

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